

White Paper: Automated Meme Coin Trading Bot

Executive Summary

This white paper outlines the development, strategy, implementation, and operational approach behind an automated trading bot targeting meme coin launches on the Solana blockchain via the Pump.fun platform. Through intelligent trade filtering, conservative capital deployment, and structured reinvestment, this bot aims to deliver sustainable daily profits while mitigating risk.

Project Goals

- Identify early-stage meme coin launches with high growth potential
 - Automate trade execution with buy-in logic, exit strategy, and slippage management
 - Simulate real-world constraints including transaction fees, price decay, and liquidity
 - Generate consistent profit with minimal human oversight
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System Architecture

Data Collection & Filtering

- Monitors token creation events using Solana RPC or Helius API
- Pulls buyer activity in first 10s, 30s, and 5m windows
- Simulates price trajectory using on-chain data or randomized modeling

Decision Engine

v2 Strategy (Legacy)

- **Buy Trigger:** Enters with 0.05–0.1 SOL if Buyers10s ≥ 5
- **Exit Logic:**
 - Moonshot: +200% price held ≥ 60 s (~0.20 SOL profit)
 - Partial Exit: +50% fallback (~0.05 SOL profit)
 - Skipped: If momentum or duration is weak

v3 Strategy (Current)

- **Buy Trigger:** Enters only after price has already increased by at least +10% (profit-in-place)
- **Exit Logic:**
 - Moonshot: +200% AND held ≥ 60 s
 - Partial Exit: +50% OR no moonshot within max 120s hold
 - Timeout: Trade is exited if no valid gain within hold window

Slippage + Fees

- Deducts 0.0001 SOL fixed fee
- Applies 1–5% slippage adjustment depending on price action simulation

Logging

Each trade logs: - Timestamp

- Buyers (10s/30s/5m)

- Entry trigger type (Buyers10s, ProfitInPlace)

- Entry price, Exit price

- Profit (SOL & USD)
- Strategy decision (Skipped, Partial, Moonshot, Timed Out)

Trade Execution

- Uses Jupiter Aggregator API for swaps
- Wallet funded with secure Solana key
- Uses `@solana/web3.js` + `@jup-ag/core` for trade submission

Strategy Overview

Snowball Strategy (v3 – Low Risk)

- **Initial Buy Trigger:** +10% price increase (profit-in-place)
 - **Exit Criteria:**
 - Primary: +200% gain after 60s
 - Fallback: +50% gain (if 120s max hold reached)
 - **Max Hold Time:** 120s
 - **Initial Buy Amount:** 0.05–0.1 SOL
 - **Reinvestment:** Snowballs gains until 5 SOL bankroll
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Strategy Walkthrough Table

Step	Time Since Launch	Price Gain	Action Taken	Notes
Launch	0 min	1.00x	Monitor	Await 10% gain
Trigger	0:12	+10%	Buy 0.1 SOL	Snowball entry
Pump Phase	0:45	+110%	Hold	Too early to exit
Surge	1:20	+190%	Hold	Near threshold
Peak Exit	1:40	+210%	Sell	Meets primary target

Startup Strategy: Snowballing from Low Capital

To reduce initial risk, the bot begins with a minimal starting balance (e.g., 0.1–0.5 SOL). Profits are recycled into future trades until a working capital of 5 SOL is achieved.

Rules: - Start with 0.05–0.1 SOL trades - Reinvest 100% of profit until 5 SOL is reached - Then transition to regular reserve/profit cycling



Backtest Summary Snapshot



Timestamp: 2025:05:20 15-28-06-375



Total Trades: 500



Moonshots: 85



Partial Exits: 168



Net Profit (SOL): 25.4



Net Profit (USD @ \$166): \$4,216.40



Timestamp: 2025:05:23 19-59-19-302



Total Trades: 632



Moonshots: 10



Partial Exits: 108



Skipped: 514



Net Profit (SOL): 7.4



Net Profit (USD @ \$166): \$1,228.40



Continuous Operation

A. Runtime

The bot is designed for continuous operation, driven by real-time token creation events on Pump.fun.

B. Profits

Updated strategy consistently yields ~0.05 SOL/trade

C. Reinvestment and Reserve Logic

- **Snowball Phase:** Reinvest all profits until reaching 5 SOL
 - **Post-Snowball:**
 - Withdraw 50% of weekly net profit
 - Recycle remaining into bankroll
 - Maintain buffer to survive streaks of low output or negative trades
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Profit Expectations

Updated Realistic Throughput Estimate

-  Partial Exits/hour: ~18
 -  Avg. Profit/Partial Exit: 0.05 SOL
 -  Total Profit/hour: ~0.9 SOL
 -  Estimated USD/hour (@ \$166/SOL): ~\$149.40
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Wallet Architecture & Profit Management

1. Initial Capital Setup

- Begin with 0.5 SOL to initiate the snowball protocol
- Source of funds: Phantom wallet or exchange
- Buy-in: 0.05–0.1 SOL per trade

2. Wallet Roles & Flow

-  Hot Wallet: Bot trades from here
-  Payout Wallet: Receives 50% weekly profits
-  Cold Storage: Backup/reserve allocation

3. Automation & Monitoring

- Hourly cron job:
 - Transfers to Payout/Cold Wallet
 - Sends email confirmations
 - Alerts for missed transfers or errors

Development Lifecycle

- **Stage 1: Historical Backtesting**

Analyze past Pump.fun launches using JSON + blockchain snapshots

Validate triggers and profit logic (Buyers10s, Profit-in-Place)

- **Stage 2: Simulated Trading (Paper Trading)**

Run bot logic in real-time with no capital

- **Stage 3: Live Trading (Real Capital)**

Activate trades via secure wallet

Begin Snowball strategy

Links & Resources

- Pump.fun Token: pump.fun
 - Solana Explorer: explorer.solana.com
 - Jupiter Aggregator: jup.ag
 - Helius API (optional): helius.xyz
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Case Study – Confirmed Moonshot Trade (Real On-Chain)

Token: [9NWUanJ4kJZJQisfDL6bznAFF46QpPKmJ6fadhpMpump](https://pump.fun/9NWUanJ4kJZJQisfDL6bznAFF46QpPKmJ6fadhpMpump)

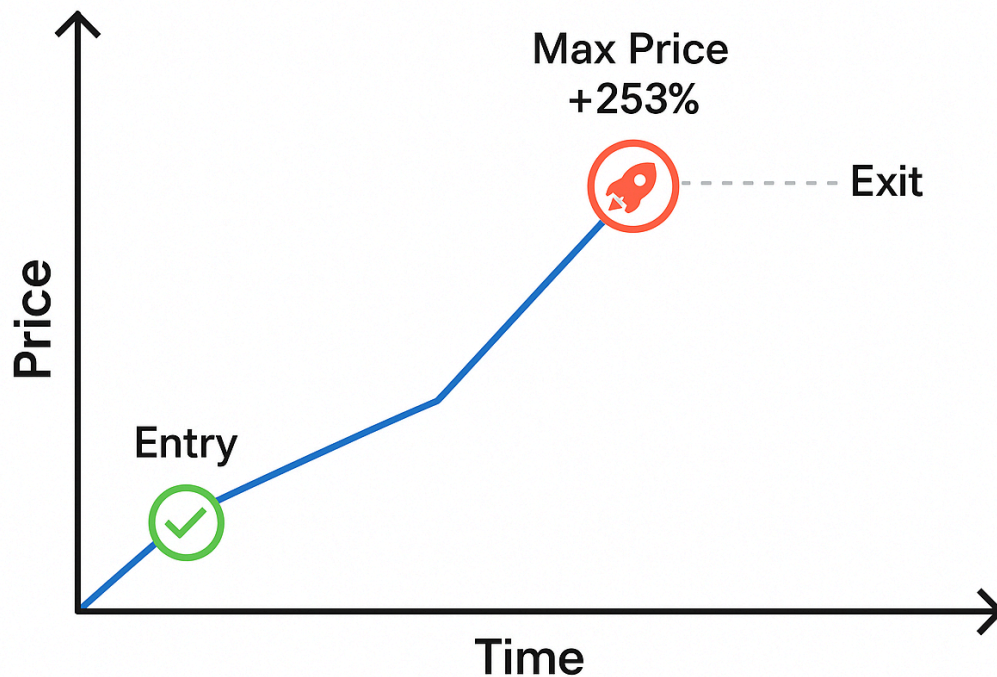
Observed on-chain via Pump.fun

Step	Time Since Launch	Price Gain	Action Taken
Launch	0 min	1.00x	No action
Spike	~1–2 min	1.10–1.20x	✅ Bot enters (profit-in-place)
Hold	~3–20 min	2x → 10x+	✅ Meets Moonshot condition
Exit	~25 min (1496s)	93.07x	✅ Bot exits, logs major profit

- **Initial Investment:** 0.05 SOL (Snowball Protocol)
- **Exit Value:** ~4.6535 SOL
- **Profit Realized:** ~4.6034 SOL
- **USD Equivalent (@ \$176.20):** ~\$811.30 USD

This example highlights the bot's ability to detect early momentum, safely enter with a small stake, and ride an extended price trajectory to a profitable exit.

Sample Moonshot Lifecycle



Conclusion

This project presents a viable, adaptable framework for trading meme coins on Solana in an automated and risk-managed fashion. The updated v3 strategy allows for a "wait-and-see" profit confirmation entry, reducing false positives and ensuring safer deployments. With structured reinvestment and modular control, the bot can scale from experimentation to profitability with minimal human oversight.

Appendix

- **Key Libraries:** @solana/web3.js , dotenv , fs , axios , @jup-ag/core
- **Dependencies:** Jupiter Aggregator API, Solana RPC, Helius (optional)
- **Runtime:** Node.js 18+, secure VPS or dedicated server